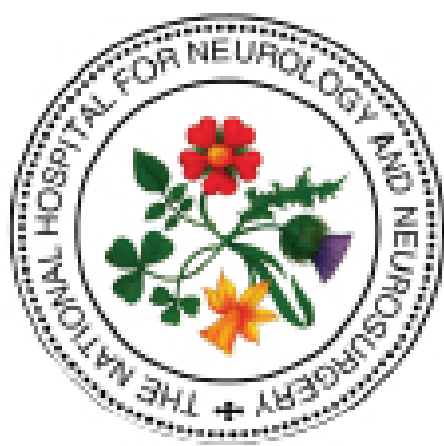


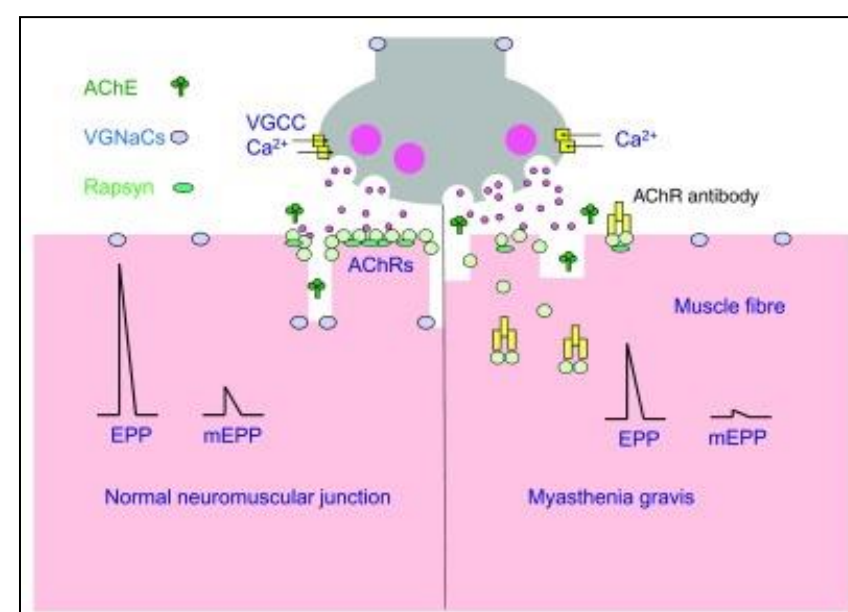


The National Hospital for Neurology and Neurosurgery

Neuroscience Critical Care Unit (NCCU)

NEUROMUSCULAR DISORDERS

MYASTHENIA GRAVIS (MG)



MG is a chronic autoimmune disease of the postsynaptic neuromuscular junction (NMJ) associated with antibodies to acetylcholine receptors. MG causes weakness in skeletal muscles which worsens after periods of activity and improves after periods of rest. The symptoms fluctuate with relapses and remissions. There is no known cure.

SIGNS and SYMPTOMS

- Ocular problems ptosis / diplopia
- Facial weakness
- Bulbar weakness—difficulty swallowing / managing saliva
- Dysarthria—voice nasal / weak / fades when talking
- Weakness / fatigability of skeletal muscle exacerbated by exercise - worsening as day goes on
- Loss of strength in limbs—arms more affected than legs / proximal muscles more affected than distal muscles
- Neck muscle weakness with head falling forward
- Respiratory muscle weakness requiring intubation / ventilation
- Bulbar dysfunction requiring intubation for airway protection

TREATMENT

No single therapy works best for all patients therefore treatment is based on an individual's response to specific therapies.

- **ANTICHOLINESTERASE** (pyridostigmine 30–90 mg every 4–6h) is the first-line treatment for symptomatic relief
- Second-line treatment includes **CORTICOSTEROIDS**: disease modifying, more effective in earlier phases of MG and should be started as low dose with gradual increase by need
- **AZATHIOPRINE** suppresses inflammation process, but response slow, so used as adjunct to steroids
- **CYCLOSPORINE, METHOTREXATE, CYCLOPHOSPHAMIDE** useful for those who do not respond to or cannot tolerate azathioprine
- **THYMECTOMY** – most patients with seropositive MG have abnormalities of thymus gland – hyperplasia or thymoma. 30–60% of thymomas associated with MG and 10% of patients with MG have thymic tumour. More common in younger patients but all screened with mediastinal imaging (CT or MRI). Surgical removal mainstay of treatment in younger patients, and adjuvant radiation for invasive thymoma recommended
- **THERAPEUTIC PLASMA EXCHANGE (TPEX)** which temporarily reduces autoantibody levels
- **INTRAVENOUS IMMUNOGLOBULIN (IVIg)** which interferes with antibody binding

RESPIRATORY MANAGEMENT in MG

The triggers used for most other patients differ in those with neuromuscular disorders.

Hypoxaemia, decreased tidal volume, dyspnoea, abnormal ABGs are all late findings in neuromuscular respiratory failure and poor indicators of the need for ventilatory support

The most sensitive indicator of progressive respiratory failure in these patients is

FORCED VITAL CAPACITY (FVC)

- **ELECTIVE INTUBATION** is considered when FVC falls to **15-20ml/kg** (normal value 75ml/kg)
- Weaning is usually commenced when FVC reaches 20ml/kg
- Bulbar dysfunction may necessitate intubation even if FVC remains adequate

DRUG REACTIONS IN MG

Several drugs (including some aminoglycoside antibiotics) exacerbate blockade at the NMJ and the use of all drugs must be carefully considered

Mg-containing antacids, laxatives, lactate-containing infusions such as Lactated Ringer's solution, should be avoided or used with caution

MAGNESIUM (Mg) 'TOP UP' – a LOWER Mg LEVEL and a HIGHER THRESHOLD FOR REPLACEMENT are INDICATED in MG

- Magnesium known to act at the NMJ by inhibiting the presynaptic release of acetylcholine and desensitizing the postsynaptic membrane. Magnesium can therefore potentiate neuromuscular weakness and has the potential to worsen symptoms and lead to a crisis.
- The role of Mg in life-threatening cardiac arrhythmias and other conditions means that strictly avoiding Mg replacement not feasible.
- **ROUTINE REPLACEMENT of Mg** should be avoided and FVC should be monitored every 2 to 4 hours after Mg replacement.

CAUTION PRE, DURING and POST-ANAESTHESIA

- Anaesthetic management of MG requires careful pre-operative assessment and optimisation to minimise post-operative complications
- Pre-operative physiotherapy and rationalization of MG treatment are necessary in all cases and, if indicated, pre-operative TPEX or IVIg should be offered
- Consistently reduced FVC and poor bulbar function are strong indicators of need for post-operative mechanical ventilation
- *ChEIs usually withheld on morning of surgery as interferes with metabolism of depolarizing / non-depolarizing NMDAs
- If on corticosteroids, give cover with hydrocortisone on induction of anaesthesia
- Patients with MG have relative resistance to the action of suxamethonium and often require twice the normal dose
- Atracurium (40–60% of normal dose) is drug of choice due to rapid, spontaneous metabolism
- Neuromuscular blocking properties of volatile anaesthetic agents can be exaggerated
- ChEIs often restarted immediately post-op, but usually the dose needs to be decreased in first 48h

REFERENCES

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ACUTE DETERIORATION in a KNOWN MG PATIENT

It is essential to distinguish between **MYASTHENIC** and **CHOLINERGIC CRISIS**

Usual cause of **MYASTHENIC CRISIS** is infection and the signs and symptoms are those of myasthenia

CHOLINERGIC CRISIS is caused by over-medication with anticholinesterase drugs and symptoms include abdominal cramping and diarrhoea (muscarinic effects) and profound generalized weakness, excessive pulmonary secretions, and impaired respiratory function (nicotinic effects)

Both are medical emergencies that require prompt tracheal intubation and assisted ventilation

